ParkinSense: An Open Wrist-Worn Research Platform for Motion, Cardiac, Activity and Sleep Signals in Parkinson's Disease Monitoring

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***Abstract*—ParkinSense is an open-source (MIT) wearable research platform that streams 104 Hz inertial and optical samples from a Seeed XIAO nRF52840 Sense board over Bluetooth Low Energy to a Python host. The host runs a tremor detector, a photoplethysmography (PPG) pipeline, a step counter, an offline sleep/wake estimator and an opt-in wear monitor. We describe the implementation in enough detail to reproduce it (wire protocol, Welch-spectrum tremor evidence with a two-state hidden Markov model, motion-gated heart-rate estimation) and label every result by the evidence behind it. On synthetic ground truth the tremor detector reaches sensitivity 0.986 and specificity 1.000 (previous version 0.692 and 0.983), at the cost of a longer onset latency (6.1 s against 1.1 s). On public wrist datasets recorded with other devices, sleep/wake agreement with polysomnography is $\kappa = 0.500$, and a gated step counter raises event F1 by +0.114 but fails two of seven pre-set criteria, so it stays opt-in. 141 automated tests pass. No patient has been recorded and no hardware measurement of accuracy, current or battery life exists; no clinical claim is made.**

Index Terms*—*Parkinson's disease, wearable sensing, tremor, photoplethysmography, step counting, sleep, Bluetooth Low Energy.

I. INTRODUCTION

Parkinson's disease (PD) produces motor signs, notably resting tremor, that vary over the day [1], [2], whereas the MDS-UPDRS [3], [4] samples them in occasional clinic visits. Wearable sensors are a proposed remedy [5]–[9], but most tremor systems analyse one inertial channel [10]–[12] and few publish the signal chain in a form others can rerun.

ParkinSense is a research prototype, not a medical device. It acquires timestamped samples, processes them with versioned, replayable algorithms, and attaches to each output an account of the evidence supporting it. The last aim is deliberate: an earlier version of this manuscript reported bench figures (current, runtime, packet loss) for which the repository holds no logs. Those statements are withdrawn here, and results are grouped into four classes: automated tests, synthetic signals, public datasets from other devices, and ParkinSense hardware (TABLE I). A result from one class is never presented as another, and no clinical class exists yet.

Contributions. (i) A reproducible description of the protocol, tremor, PPG, step, sleep and wear algorithms. (ii) A before/after benchmark on identical synthetic input. (iii) Participant-independent evaluations on public wrist data, including a candidate that failed its pre-set acceptance rule. (iv) An explicit statement of what remains unmeasured on the device.

TABLE I
EVIDENCE CLASSES AND WHAT EACH CAN SUPPORT.

Class	Source	Supports
Automated test	pytest, host firmware tests, CI	code behaves as specified on stubbed or synthetic input
Synthetic	seeded generators with known truth	relative behaviour and regressions, not accuracy on people
Public data	other devices (Apple Watch, ActiGraph, Shimmer)	behaviour on real wrists, not on this hardware
ParkinSense hardware	logged recordings and measurements	device-level claims (rate, loss, current, wear)
Clinical	labelled patient data, ethics approval	**none exists**

II. RELATED WORK

Resting PD tremor lies at about 4–6 Hz within a 3.5–7.5 Hz search band [13] and is separable from voluntary motion with spectral features of wrist or finger inertial signals [10], [12]; classifiers have been used to distinguish it from essential tremor [11], [14]. Amplitude depends on posture and activity [15], which motivates reporting motion context with each reading. ParaDigMa [16] is an open pipeline that windows wrist gyroscope data into 4 s segments and classifies them with an MFCC-based model (published leave-one-subject-out sensitivity 0.61, specificity 0.97); ParkinSense borrows its aggregation definitions but not its classifier, and has not been run on its data. ELENA [17], [18] is a low-cost finger-worn MPU6050/ESP32 device with offline spectral analysis, validated on a rig and a smartwatch and in a nine-patient pilot; ParkinSense has no comparable study. Wrist PPG is established for heart rate but degraded by motion [19], [20]; step counting [21], free-living activity [22], [23] and accelerometer sleep/wake [24] are mature topics, and ParkinSense implements small auditable versions benchmarked against published baselines. TABLE II summarises the positioning.

TABLE II
POSITIONING OF RELATED SYSTEMS (AS PUBLISHED).

System	Sensing and processing	Evidence reported
Inertial tremor studies [10], [12]	wrist or finger accelerometer/gyroscope; offline features	small cohorts
ELENA [17], [18]	finger MPU6050, ESP32; offline MATLAB FFT	rig, smartwatch, 9 PD + 9 controls
ParaDigMa [16]	wrist gyroscope; open 4 s-window MFCC model	LOSO sens. 0.61, spec. 0.97
ParkinSense	IMU + finger PPG over BLE; host-side, replayable	synthetic and public data; **no patients**

III. SYSTEM, FIRMWARE AND DATA INTEGRITY

Architecture. Sensing (LSM6DS3 inertial sensor, MAX30102 optical sensor on a 400 kHz I²C bus, Seeed XIAO nRF52840 Sense with a 3.7 V Li-ion cell) feeds firmware that only samples, timestamps and packs data (Fig. 1). All processing is on the host, so algorithms change without reflashing and recordings can be reanalysed. Processing uses *signal time* from device timestamps (the 32-bit microsecond counter, which wraps every 71.6 min, is unwrapped), so a recording replayed at any speed reproduces the live output; the regression suite relies on this.

Firmware. Version 4 (protocol v1) is the last build verified on hardware. Version 5.0.0 (protocol v2) fixes defects found in audit, listed in TABLE III; it has been tested on a host and type-checked against API stubs, but *never compiled for the target, flashed or measured*. A scheduler runs at 9 615 µs (104.004 Hz) and sets a discontinuity flag after more than five missed intervals. In the shipped profile the device streams only while a finger keeps the infrared level above 50 000 counts (three consecutive 250 ms polls); without finger contact there is no inertial stream. An alternative profile with independent sensors type-checks but is untested.

TABLE III
FIRMWARE V4 AND 5.0.0 (HOST-TESTED ONLY).

Item	v4 (protocol v1)	5.0.0 (protocol v2)
IMU range, rate	±16 g, ±2000 dps (intended range never applied), 416 Hz sensor	±4 g, ±500 dps, 104 Hz; one 12-byte burst read
PPG to host	10 Hz (one sample per 100 ms)	≈ 100 Hz
Over-range, faults	wraps; not reported	saturates; 7 header flags
Packet	208 B: 8 B header + 10 × 20 B samples	same; sequence number added
Status	**hardware-verified**	**not flashed**

Protocol and integrity. Both versions send 208-byte notifications (≈ 10.4 packets/s) of ten samples (ax, ay, az, gx, gy, gz, IR, red); accelerometer scale is 8192 counts/g and the gyroscope scale is 131 (v1) or 64 counts/dps (v2), selected by a version field. Version 2 adds a sequence number and flags for clipping, IMU/PPG fault, stale PPG, failed notify and time discontinuity. The host decoder is the only parser and re-encodes byte-exactly. The true stream rate is measured from timestamps over 100 packets and classed OK within 2 %, WARN within 5 % and INVALID beyond, since spectral frequencies scale with rate error. A gap longer than 0.5 s resets every analysis window. Session metadata records all versions and counters.

Software. The host package (Python ≥ 3.10; NumPy, SciPy, pandas) separates protocol, motion, tremor, PPG, activity, sleep and wear modules from the runtime, which exposes a bounded FastAPI endpoint. A Plotly Dash dashboard consumes it, shows a SIMULATED banner whenever data do not come from the wearable, labels values below VALID quality as not measurements, and replays recorded sessions through the same rendering path. Capture files (79 columns) record the device, host-monotonic and host-wall clocks.

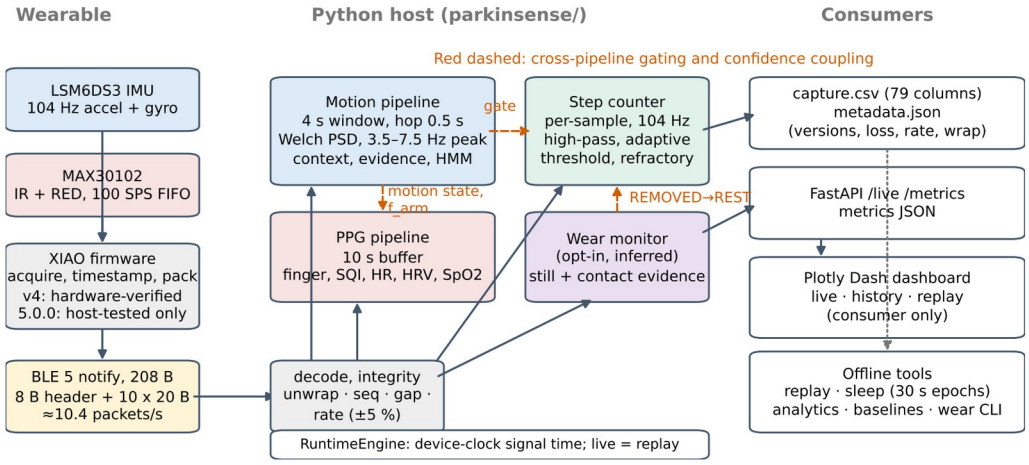


Fig. 1. ParkinSense architecture. Solid arrows: data flow; dashed red arrows: gating and confidence couplings between pipelines.

IV. SIGNAL-PROCESSING ALGORITHMS

A. Tremor detection

Every 0.5 s the pipeline analyses a 4 s window (416 samples). Each gyroscope axis gets a Welch periodogram [25] (2 s Hann segments, 50 % overlap, K = 3, linear detrend, FFT length 1024); resolution is set by the segment (0.5 Hz bin, ≈ 0.72 Hz Hann main lobe) and zero-padding to 0.1016 Hz only interpolates. The axis-summed spectrum is

$$S(f)=\sum_{a\in\{x,y,z\}} P_a(f) \tag{1}$$

with P_a the Welch estimate of axis a. The gyroscope is intentionally not band-passed first, because an earlier band-passed version pushed the band ratio towards one for almost any input. The peak of S in 3.5–7.5 Hz is refined by Gaussian interpolation [26], where α, β, γ are the log-spectrum at the peak bin and its neighbours:

$$\hat{f}=f_{i^*}+\frac{\alpha-\gamma}{2(\alpha-2\beta+\gamma)}\Delta f \tag{2}$$

Tremor power is the integral of S within ±0.625 Hz of the peak, giving an amplitude in °/s RMS [27], and the noise floor F is the median of S over 2–12 Hz excluding the peak ±1 Hz:

$$A_T=\sqrt{\sum_{|f_i-\hat{f}|\le 0.625} S(f_i)\,\Delta f}\,,\quad SNR=10\log_{10}\frac{S(f_{i^*})}{F} \tag{3}$$

Voluntary movement is measured in the time domain: A_{arm} is the RMS of the 0.5–3 Hz band-passed gyroscope, and gravity-removed acceleration is band-limited to 0.2–3 Hz. Thresholds of 0.03/0.20 g and 10/30 °/s give REST, LOW MOTION or ACTIVE. Window evidence combines logistic gates on SNR and amplitude, rejects peaks within 0.15 Hz of the band edges, and down-weights arm-swing harmonics (h = 0.15 at 2–4 times the arm frequency, else 1):

$$e=\sigma\left(\frac{SNR-10}{2}\right)\sigma\left(\frac{A_T-1}{0.25}\right)1\left[3.65<\hat{f}<7.35\right)h \tag{4}$$

Consecutive windows share 87.5 % of their samples, so evidence is not independent. A two-state hidden Markov model [28] tempers each likelihood ratio by the elapsed signal time Δt (ε is a small constant), with time constants τ_{in} = 120 s, τ_{out} = 60 s, Λ_{max} = 60, ε = 10^{-3} and prior 0.02:

$$p^- = p\,e^{-\Delta t/\tau_{out}} + (1-p)\left(1-e^{-\Delta t/\tau_{in}}\right) \tag{5}$$

$$p' = \frac{\Lambda\, p^-}{\Lambda\, p^- + 1 - p^-} \tag{6}$$

$$\Lambda = clip\left(r,\Lambda_{max}^{-1},\Lambda_{max}\right)^{min(1,\Delta t/4)},\quad r=\frac{e+\varepsilon}{1-e+\varepsilon} \tag{7}$$

Tremor is published at $p \ge 0.8$ and released at ≤ 0.2. With full evidence, the first window gives p = 0.763 and the next 0.5 s hop

0.839, so one 4 s burst cannot confirm alone. Fig. 2 and Fig. 3 show spectra of four synthetic scenarios and the resulting evidence; on a rest–tremor–rest session the state lags onset by 6.0 s and offset by 8.0 s. TABLE IV lists the settings. The output has no severity score. Physiological tremor at 6–7.5 Hz with enough amplitude is reported as tremor, and the detector has not been evaluated on people with PD.

TABLE IV
PRINCIPAL PARAMETERS (ENGINEERING SETTINGS, NOT FITTED TO PATIENT DATA).

Pipeline	Setting
Tremor	4 s window, 0.5 s hop; search 3.5–7.5 Hz; gates 10 dB and 1.0 °/s; context 0.03/0.20 g, 10/30 °/s; HMM τ_{in} 120 s, τ_{out} 60 s, confirm 0.8, release 0.2
PPG	10 s buffer; band 0.5–4.5 Hz; RR 45–180 bpm; EMA 0.25; coupling tolerance 0.12 Hz; HRV above 25 Hz only; SpO_2 clipped to 70–100 %
Steps	high-pass $\alpha = 0.95$; threshold $\mu + 0.7\sigma$ over 2 s; refractory 0.30 s; walking after 3 steps; gated: $\rho \geq 0.65$, RMS ≥ 0.12 g
Sleep, wear	30 s epochs, $\theta = 0.7$; stillness 1 mg SD; possible/removed after 300/900 s; contact IR > 50 000

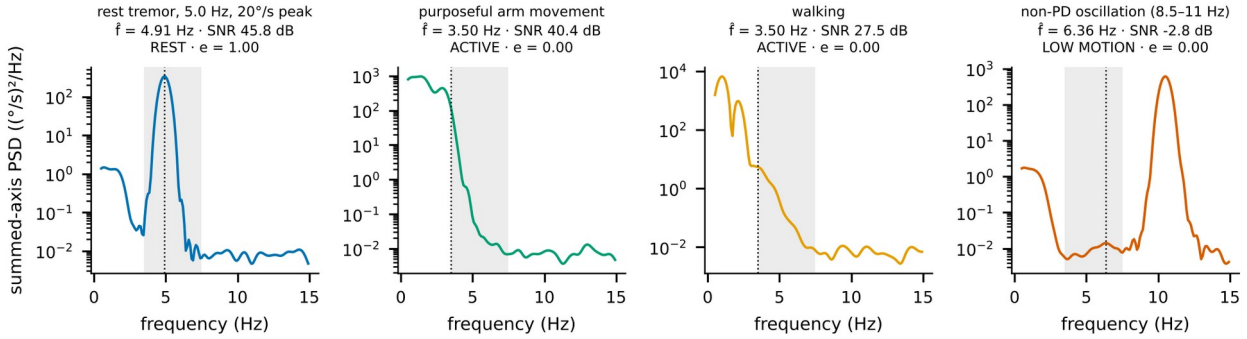


Fig. 2. Summed-axis gyroscope spectra of four synthetic scenarios (search band shaded, interpolated peak dotted). Titles give peak, SNR, context and evidence *e*.

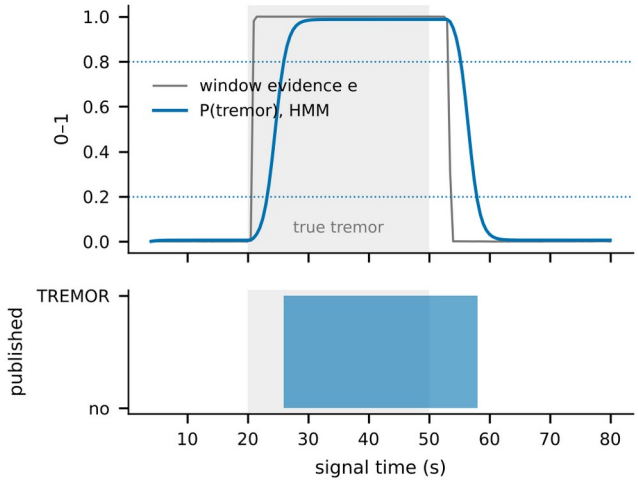


Fig. 3. Window evidence and HMM posterior on a synthetic 20 s rest / 30 s tremor / 30 s rest session through the full runtime.

B. Photoplethysmography

A 10 s buffer is analysed every 0.5 s. Finger presence uses seven weighted checks with 1 s (on) and 2 s (off) hysteresis, and a 0–100 signal-quality index combines nine sub-scores. Heart rate uses a deliberately simple chain: a 0.5–4.5 Hz third-order zero-phase Butterworth filter, peak detection (0.30 s minimum spacing, prominence ≥ 0.5 SD, dicrotic-notch guard), RR intervals within 45–180 bpm, and the median interval, smoothed by an exponential average:

$$HR = \frac{60}{median\left(r_k\right)}, h_t = 0.25\, HR_t + 0.75\, h_{t-1} \quad (8)$$

Rhythmic arm motion can capture the beat detector; the pipeline *flags* heart rate within 0.12 Hz of the arm-movement frequency or its second harmonic and scales confidence by 1.0/0.8/0.5 for REST/LOW MOTION/ACTIVE and by 0.25 when coupled. Unlike TROIKA [20], it does not correct the artefact. Heart-rate variability follows standard definitions [29] but is withheld below 25 Hz effective PPG rate, which includes every firmware-v4 stream (10 Hz). SpO_2 uses a generic ratio-of-ratios curve and is an *uncalibrated estimate*; no reference oximeter data exist:

$$SpO_2 = 97.601 + 10.503\, R - 11.266\, R^2, R = \frac{AC_{red}/DC_{red}}{AC_{IR}/DC_{IR}} \quad (9)$$

C. Steps, sleep and wear

Steps. The default counter high-passes the accelerometer magnitude *m* (corner ≈ 0.83 Hz) and compares it with an adaptive threshold over the last 2 s:

$$y_n = 0.95\left(y_{n-1} + m_n - m_{n-1}\right), T = clip\left(\mu_{2s} + 0.7\,\sigma_{2s}, 0.08, 1.5\right)g \quad (10)$$

A step is an upward crossing when armed, not at REST, and at least 0.30 s after the last; a confirmed rest tremor forces the gate to REST (the *tremor guard*). An opt-in gated counter (0.5–3.5 Hz band-pass, autocorrelation ≥ 0.65 over lags 0.30–1.40 s, 2 s RMS ≥ 0.12 g [30]) counts candidate peaks only while a walking gate is open. **Sleep.** An offline estimator forms 30 s epochs from six arm-angle and ENMO context features [24] and applies an L2 logistic regression (IRLS) with a threshold of 0.7 chosen by nested leave-one-subject-out; it gives no stages and is not validated in PD. **Wear.** An opt-in host-side monitor changes neither firmware nor protocol and declares removal only after 300 s (possible) or 900 s (removed) of continuous stillness *and* absent optical contact; stillness alone, contact loss alone and data gaps never imply removal, and re-wear by motion cannot restore the WORN state without contact (Fig. 4). **Baselines.** Session

summaries follow ParaDigMa [16]; a personal baseline flags deviation from a 28-day rolling median at $|z| \geq 3$ (MAD scale) and has no diagnostic meaning. Bradykinesia, dyskinesia and freezing of gait are not implemented.

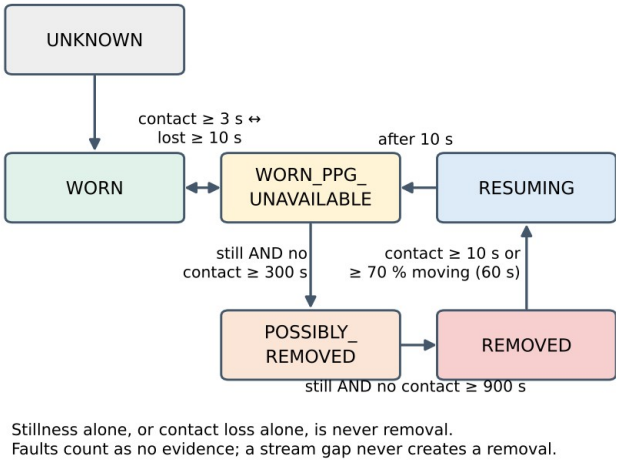


Fig. 4. Wear-state model (simplified). REMOVED needs continuous stillness and absent optical contact; re-wear by motion lands in WORN_PPG_UNAVAILABLE.

V. EVALUATION METHODOLOGY

Synthetic. Seeded generators produce rest and action tremor (4–6.5 Hz with frequency and amplitude modulation), voluntary movement, walking, vibration, an 8.5–11 Hz oscillation and a two-Gaussian PPG model with optional motion artefact. A harness feeds byte-identical packets to the repository as originally found (pipeline 2.2.0) and to the current one (16 seeds × 30 s per tremor scenario, first 8 s excluded). Results show relative behaviour only. **Public data.** Sleep: Walch [31], 17 subjects, leave-one-subject-out, paired subject-level bootstrap [32]. Steps: Clemson [33] (30 participants, 54 374 foot contacts, 15 Hz wrist data up-sampled to 104 Hz) and Karas [34] (27 subjects, 641 min of driving and clapping, no step events). Wear: Walch nights and Karas driving. These are other devices on other wrists. **Rules.** Acceptance criteria and hardware pass criteria were written before any candidate or measurement existed; thresholds were chosen only inside participant-independent loops; decisions that used evaluation data are reported as optimistic. **Tests.** 141 pytest tests pass (1 deselected, no board) in the run of 2026-11-08 (Linux x86-64, Python 3.13); TABLE V lists the test layers. Host compute is 26.4× real time with a 17.8 ms median per analysis packet; this is not BLE or end-to-end latency.

TABLE V
SOFTWARE TEST LAYERS AND STATUS.

Layer	Content	Result
pytest	unit, integration, golden regression, synthetic floors	**141 passed**, 1 deselected (hardware)
flake8	CI rule set over package, tests, benchmarks	0 errors
Firmware host tests	struct sizes, scaling, saturation, flags, scheduler with timer wrap	OK
Sketch type-check	build profiles A and B against API stubs	OK; does not prove an nRF52 build
Golden replay	6 real recordings, 104 Hz BLE and ≈ 23 Hz serial logs	unchanged by default pipeline
Hardware tests	H, W and PPK2 series (Section VII)	**NOT RUN**

VI. RESULTS

TABLE VI
TREMOR BENCHMARK ON SYNTHETIC SIGNALS (16 SEEDS): EARLIER (2.2.0) AND CURRENT (3.0.0) PIPELINES.

Metric	2.2.0	3.0.0
Sensitivity / specificity	0.692 / 0.983	**0.986 / 1.000**
F1; TP / FN / FP / TN	0.798; 997/443/62/3538	**0.993**; 1420/20/0/3600
120–250 °/s tremor detected	38 %	97 %
8.5–11 Hz called tremor	8.6 %	0.0 %
Action tremor / walking + tremor detected	49 % / 60 %	100 % / 100 %
Frequency error, median / p90 (Hz)	0.090 / 0.219	0.024 / 0.096
Onset / offset latency (s)	1.1 / 4.0	6.1 / 8.1

TABLE VI shows that the new detector removes three failure modes of its predecessor: strong tremor being vetoed as voluntary movement (38 % → 97 % detected), an 8.5–11 Hz oscillation being called PD tremor (8.6 % → 0 %), and tremor during action or walking being missed (49 %, 60 % → about 100 %). Latency rises about fivefold by design (Fig. 5). These are synthetic results with hand-set thresholds, not clinical accuracy.

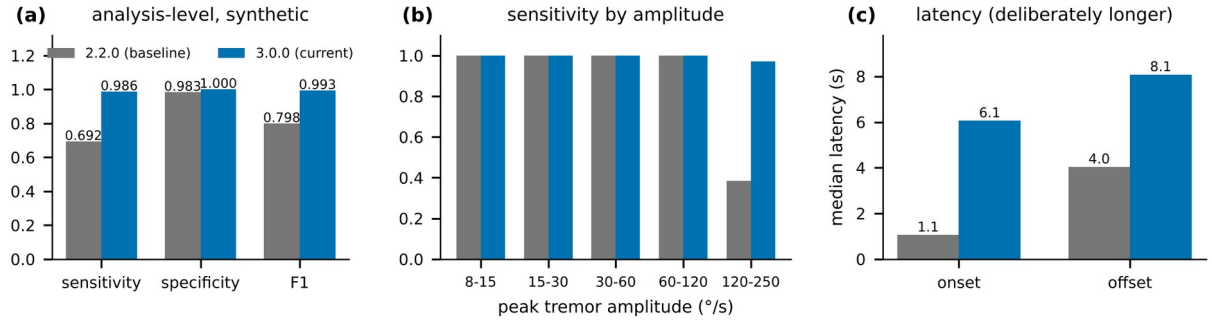


Fig. 5. Synthetic tremor benchmark: (a) analysis-level metrics; (b) sensitivity by tremor amplitude; (c) median onset and offset latency.

TABLE VII
PPG ON SYNTHETIC SIGNALS (REST: 10 SEEDS × 60 S; ARTEFACT: 6 SEEDS, 1.8 HZ). V1/V2: FIRMWARE-V4/5.0.0 DATA PATHS.

Metric	v1 2.2.0	v1 3.0.0	v2 2.2.0	v2 3.0.0
HR MAE at rest (bpm)	16.1	0.29	17.8	0.37
RMSSD MAE at rest (ms)	195	withheld	91.6	6.7
HR confidence: rest / artefact	0.44 / 0.63	0.59 / 0.14	0.49 / 0.84	0.73 / 0.17
HR MAE under artefact (bpm)	29.5	27.7	27.9	28.0

In TABLE VII and Fig. 7, the resting heart-rate gain (16–18 → 0.3–0.4 bpm) comes from correcting a sample-rate mismatch (50 Hz assumed, 104 Hz received). Under artefact the error is unchanged (≈ 28 bpm) but now flagged: confidence falls from 0.84 to 0.17. On synthetic steps the walking error is −9.7 % for both versions, while false steps per minute fall from 23.3 to 0.3 during rest tremor (Fig. 6) and are unchanged during purposeful arm movement (46.3 → 48.8).

The default counter overcounts irregular gait on Clemson. The gated counter raises held-out event F1 from 0.639 to 0.753 (paired bootstrap Δ = +0.114, 95 % CI [+0.092, +0.134]) and reduces false steps on Karas, at lower walking sensitivity (TABLE VIII, Fig. 8). Of seven pre-set criteria it fails two (no condition worse by more than 2 MAPE points: SemiRegular 20.9 against 8.2 %; walking sensitivity within 2 points: 0.900 against 0.973), so production v1.1 remains the default. The Karas runtime motion gate was held ACTIVE, so its false-step rate probably overstates the full pipeline.

TABLE VIII
STEP COUNTING ON PUBLIC WRIST DATA. CLEMSON: BIAS % / MAPE % / EVENT F1 (GATED IS HELD OUT BY 2-FOLD PARTICIPANT SPLIT); KARAS: FALSE STEPS PER MINUTE AND WALKING SENSITIVITY.

Counter	Clemson all	Clemson irregular	Karas FP/min; sens.
Production v1.1	+13.3 / 111.1 / 0.639	+299.3 / 303.4 / 0.297	52.5; 0.973
Periodic (experimental)	-37.2 / 41.3 / 0.387	-47.5 / 47.2 / 0.178	2.4; 0.509
Gated 3.0-dev	-9.7 / 18.2 / 0.753	+10.2 / 26.3 / 0.569	4.1; 0.900

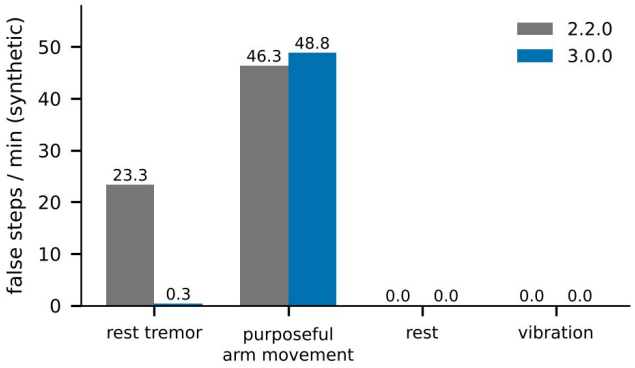


Fig. 6. False steps per minute on synthetic scenarios for the earlier and current runtime fed identical packets: the tremor guard removes false steps during rest tremor.

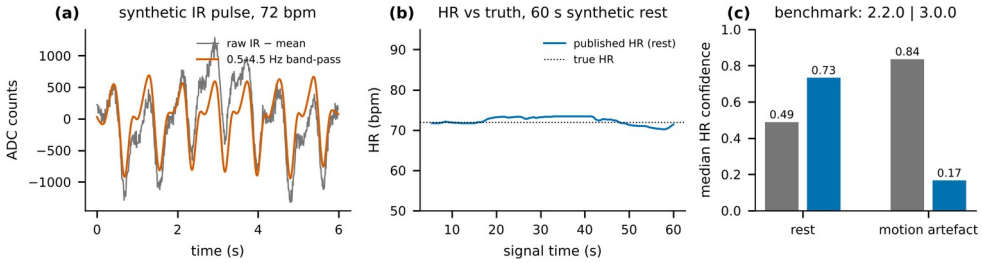


Fig. 7. PPG on synthetic signals: (a) IR pulse and band-passed trace; (b) heart rate versus truth at rest; (c) median confidence at rest and under artefact.

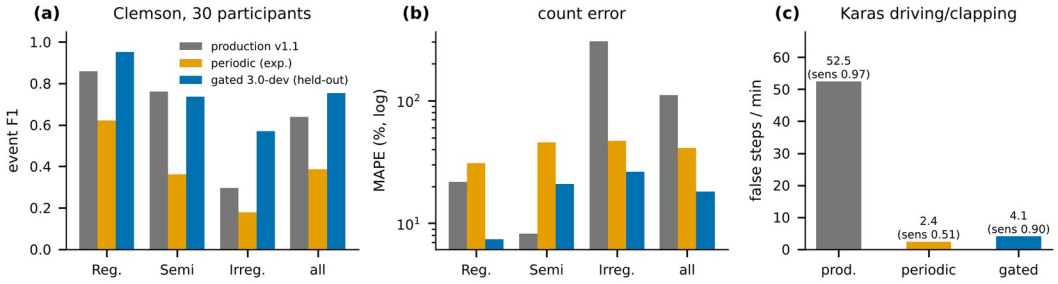


Fig. 8. Step counters on public wrist data: (a) event F1 and (b) count error by gait condition on Clemson; (c) false steps per minute on Karas.

TABLE IX
SLEEP/WAKE ON WALCH (13,674 EPOCHS, 17 SUBJECTS, 90.5 % SLEEP).

Method	κ	Bal. acc.	Wake spec.	TST MAE (bias), min
van Hees z-angle rule [24]	0.414	0.765	0.618	21.3 (-13.9)
Logistic regression (shipped)	0.500	0.767	0.581	14.0 (+1.3)
LR + tempered HMM	0.472	0.729	0.492	14.0 (+8.2)
LR + heart rate + HMM	0.463	0.719	0.472	14.1 (+8.7)

In TABLE IX and Fig. 9 the shipped model agrees better with polysomnography than the heuristic (Δκ = +0.086, 95 % CI [+0.047, +0.127], 14/17 subjects better) and total sleep time better (|error| -7.3 min), while balanced accuracy is unchanged and wake detection stays weak. Seventeen healthy adults and one scorer limit generalisation. For wear, in 15 worn Walch nights 36.2 % of the night lies in still runs of ≥ 15 min at the 13 mg non-wear threshold, so motion alone cannot decide removal. With optical contact forced absent (worst case), the 1 mg default marks no night REMOVED, against 5/15 at 2 mg and 12/15 at 4 mg, and 0.0 % of 623 min of Karas driving. The threshold was tuned on these nights, so this is optimistic, and sensitivity to real removal is unmeasured. TABLE X gives the wear numbers. Unlabelled real recordings (TABLE XI) are used only for regression; a wrist shaken at 8–9 Hz is correctly not called PD-band tremor, whereas arm movement and shaking add false steps in the legacy-rate logs.

TABLE X
WEAR INFERENCE ON PUBLIC DATA, OPTICAL CONTACT FORCED ABSENT (WORST CASE).

Stillness threshold	Worn night in still runs ≥ 15 min	Mean REMOVED	Subjects ever REMOVED
1 mg (default)	0.0 %	0.0 %	0 / 15
2 mg	3.3 %	6.7 %	5 / 15
4 mg	13.2 %	42.5 %	12 / 15

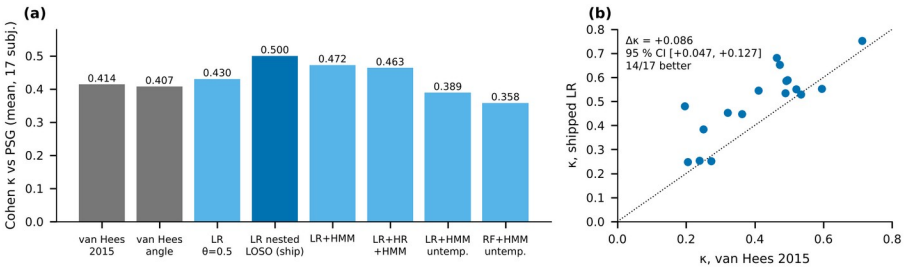


Fig. 9. Sleep/wake versus polysomnography on Walch: (a) κ by method; (b) per-subject κ of the shipped model against the published heuristic.

TABLE XI
UNLABELLED REAL RECORDINGS: GOLDEN REGRESSION OUTPUTS (GUARD AGAINST REGRESSION; NOT ACCURACY).

Recording	Tremor	Steps	Motion state
stationary, 60 s	0 %	0	REST 100 %
walking, 60 s	0 %	69	ACTIVE 100 %
everyday arm movement, 60 s	0 %	90	ACTIVE 95 %, LOW MOTION 5 %
wrist shaken at 8–9 Hz, 30 s	0 %	83	LOW MOTION 51 %, ACTIVE 49 %
vibration, 30 s	0 %	3	LOW MOTION 54 %, REST 46 %

VII. HARDWARE STATUS, LIMITATIONS AND FUTURE WORK

Hardware. Every documented hardware test is NOT RUN: board wiring (W0), firmware-5.0.0 streaming and sequence continuity (H1–H3), current with a power analyser (S1–S7), bench and worn-still noise (W1–W2), removal and re-wear (W3–W4), sensor independence (W5–W7) and an overnight recording (W8). The owner reported on 2026-11-06 that updated firmware and dashboard ran on the owner; with no firmware version, duration, loss figure or file, this raises confidence that the device streams but satisfies no pass criterion. The last documented hardware evidence is an 87 s firmware-v4 capture at 104.01 Hz (TABLE XII; earlier claims in TABLE XIII). Current, battery life, packet loss over long sessions, wrist PPG and overnight operation are therefore all unmeasured.

TABLE XII
HARDWARE TESTS AND STATUS (PASS CRITERIA ARE FIXED IN THE VALIDATION PLAN).

ID	Test	Status
W0	record wiring and MAX30102 interrupt line (board: XIAO nRF52840)	NOT RUN
H1–H3	firmware-5.0.0 init, streaming, sequence continuity, loss	NOT RUN
S1–S7	current with a power analyser	NOT RUN
W1–W2	bench noise floor; worn-still distribution (≥ 5 nights, ≥ 3 people)	NOT RUN
W3–W4	removal and re-wear detection	NOT RUN
W5–W8	sensor independence, wrist optical signal, overnight ≥ 8 h	NOT RUN

TABLE XIII
STATEMENTS OF THE EARLIER MANUSCRIPT THAT ARE WITHDRAWN OR RESTATED.

Earlier statement	Treatment
≈ 7 mA, ≈ 100 h runtime, µA deep sleep	withdrawn; no current measurement, v4 has no sleep code
no packet loss over multi-hour sessions	withdrawn; no log, loss counters exist only from v2
< 100 ms latency after window close	restated as host compute time (17.8 ms median)
bench tables “measured/validated”	withdrawn; no logs; replaced by evidence classes
composite score, 5-of-10 window rule, z-score PPG detector	replaced by evidence + HMM and the simple HR chain
wrist PPG; superiority over a published clinical pilot	corrected (finger sensor); comparison removed

Discussion. The largest gains in this revision came from correcting plumbing rather than inventing algorithms: a 2.08× sample-rate mismatch in the PPG chain, a motion multiplier that was never applied, a state machine whose dead band froze its earlier state, and host-arrival timestamps that quantised intervals. Measuring the true stream rate, carrying versions with every output and testing live processing against replay made these defects visible. The stability/latency trade-off is explicit: a 6 s onset suits a longitudinal biomarker stream, but not a trigger for an actuator. Relative to ELENA [17] the platform has no clinical pilot, and relative to ParaDigMa [16] no free-living evaluation; its contribution is an open, versioned and replayable stack whose inertial algorithm can be run on the same public data once access is confirmed. Whether its additional channels improve any patient-level outcome is untested.

Limitations. There are no patient data and no clinical claim. The inertial stream exists only with finger contact on the optical sensor in the shipped profile. Tremor evidence is not a calibrated probability, physiological tremor can be reported as tremor, and onset latency is 6.1 s, acceptable for longitudinal biomarkers but not for an actuator. SpO_2 is uncalibrated; HRV is a 10 s trend value; heart rate under motion is flagged rather than corrected. The default step counter overcounts irregular gait and arm movement. Sleep and wear evidence come from healthy adults and other devices.

Future work. The first step is one documented hardware session (W0, H1–H3, W1–W2, and a current subset) because W1 and W2 decide whether an extra skin-contact sensor is needed. Next are public tremor and freezing-of-gait baselines (ParaDigMa data and Daphnet, access to be confirmed), then a mobile application and, only under ethics approval, a patient pilot.

VIII. CONCLUSION

ParkinSense now has a specified, versioned and testable implementation: a 104 Hz protocol with fault flags, a motion-gated tremor detector with temporal inference, an auditable heart-rate pipeline, a step counter whose better candidate remains opt-in, and offline sleep and wear estimators. On synthetic and public data these behave as described; what they do on the device and on people with Parkinson’s disease is not yet known, and a documented hardware session and an ethics-approved clinical study are the next steps.

DATA AND CODE AVAILABILITY

github.com/sudo-pranshu/ParkinSense (MIT). Tests: `pytest`. Benchmarks and committed results: `benchmarks/`. Public datasets are fetched by `benchmarks/external/fetch.py` and not redistributed. Paper sources: `docs/research/paper/`.

REFERENCES

[1] J. Jankovic, "Parkinson's disease: clinical features and diagnosis," *J. Neurol. Neurosurg. Psychiatry*, vol. 79, pp. 368–376, 2008.

[2] Parkinson's Foundation, "Tremor." [Online]. Available: https://www.parkinson.org/understanding-parkinsons/movement-symptoms/tremor

[3] C. G. Goetz *et al.*, "Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): process, format, and clinimetric testing plan," *Mov. Disord.*, vol. 22, pp. 41–47, 2007.

[4] C. G. Goetz *et al.*, "Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results," *Mov. Disord.*, vol. 23, pp. 2129–2170, 2008.

[5] C. F. Pasluosta, H. Gassner, J. Winkler, J. Klucken, and B. M. Eskofier, "An emerging era in the management of Parkinson's disease: wearable technologies and the Internet of Things," *IEEE J. Biomed. Health Inform.*, vol. 19, pp. 1873–1881, 2015.

[6] W. Maetzler, J. Domingos, K. Srulijes, J. J. Ferreira, and B. R. Bloem, "Quantitative wearable sensors for objective assessment of Parkinson's disease," *Mov. Disord.*, vol. 28, no. 12, pp. 1628–1637, 2013.

[7] E. Rovini, C. Maremmani, and F. Cavallo, "How wearable sensors can support Parkinson's disease diagnosis and treatment: a systematic review," *Front. Neurosci.*, vol. 11, art. 555, 2017.

[8] A. J. Espay *et al.*, "Technology in Parkinson's disease: challenges and opportunities," *Mov. Disord.*, vol. 31, no. 9, pp. 1272–1282, 2016.

[9] A. J. Espay *et al.*, "A roadmap for implementation of patient-centered digital outcome measures in Parkinson's disease obtained using mobile health technologies," *Mov. Disord.*, vol. 34, pp. 657–663, 2019.

[10] G. Rigas *et al.*, "Assessment of tremor activity in the Parkinson's disease using a set of wearable sensors," *IEEE Trans. Inf. Technol. Biomed.*, vol. 16, pp. 478–487, 2012.

[11] O. Bazgir, S. A. H. Habibi, L. Palma, P. Pierleoni, and S. Nafees, "A classification system for assessment and home monitoring of tremor in patients with Parkinson's disease," *J. Med. Signals Sens.*, vol. 8, pp. 65–72, 2018.

[12] H. Dai, P. Zhang, and T. C. Lueth, "Quantitative assessment of parkinsonian tremor based on an inertial measurement unit," *Sensors*, vol. 15, pp. 25055–25071, 2015.

[13] G. Deuschl, P. Bain, and M. Brin, "Consensus statement of the Movement Disorder Society on tremor," *Mov. Disord.*, vol. 13, suppl. 3, pp. 2–23, 1998.

[14] Y. Zhang *et al.*, "Tremor-based discrimination of Parkinson's disease and essential tremor using a wearable sensor: a comparative study," *Sci. Rep.*, vol. 12, art. 4318, 2022.

[15] J. Henderson, C. Yiannikas, J. Morris, R. Einstein, D. Jackson, and K. Byth, "Postural tremor of Parkinson's disease," *Clin. Neuropharmacol.*, vol. 17, pp. 277–285, 1994.

[16] N. A. Timmermans *et al.*, "A generalizable and open-source algorithm for real-life monitoring of tremor in Parkinson's disease," *npj Parkinson's Dis.*, 2025 (PMC12246068).

[17] Y. Saez, C. Ureña, J. Valenzuela, A. García, and E. Collado, "A wearable Internet of Things-based device for the quantitative assessment of hand tremors in Parkinson's disease: the ELENA project," *Sensors*, vol. 25, no. 9, art. 2763, 2025, doi: 10.3390/s25092763.

[18] L. Peña Batista, C. Ureña, E. Collado, and Y. Saez, "Initial validation of ELENA: an affordable device for monitoring tremors in Parkinson's patients," in *Proc. LACCEI*, 2024, doi: 10.18687/LACCEI2024.1.1.1568.

[19] T. Tamura, Y. Maeda, M. Sekine, and M. Yoshida, "Wearable photoplethysmographic sensors—past and present," *Electronics*, vol. 3, no. 2, pp. 282–302, 2014.

[20] Z. Zhang, Z. Pi, and B. Liu, "TROIKA: a general framework for heart rate monitoring using wrist-type photoplethysmographic signals during intensive physical exercise," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 2, pp. 522–531, 2015.

[21] C.-C. Yang and Y.-L. Hsu, "A review of accelerometry-based wearable motion detectors for physical activity monitoring," *Sensors*, vol. 10, no. 8, pp. 7772–7788, 2010.

[22] J. L. Adams *et al.*, "Multiple wearable sensors in Parkinson and Huntington disease individuals: a pilot study in clinic and at home," *Digit. Biomark.*, vol. 1, pp. 52–63, 2017.

[23] J. L. Adams *et al.*, "A real-world study of wearable sensors in Parkinson's disease," *npj Parkinsons Dis.*, vol. 7, art. 106, 2021.

[24] V. T. van Hees *et al.*, "Estimating sleep parameters using an accelerometer without sleep diary," *Sci. Rep.*, vol. 8, art. 12975, 2018; and V. T. van Hees *et al.*, "A novel, open access method to assess sleep duration using a wrist-worn accelerometer," *PLoS ONE*, vol. 10, no. 11, art. e0142533, 2015.

[25] P. D. Welch, "The use of fast Fourier transform for the estimation of power spectra," *IEEE Trans. Audio Electroacoust.*, vol. 15, no. 2, pp. 70–73, 1967.

[26] E. Jacobsen and P. Kootsookos, "Fast, accurate frequency estimators," *IEEE Signal Process. Mag.*, vol. 24, no. 3, pp. 123–125, 2007.

[27] R. J. Elble and J. McNames, "Using portable transducers to measure tremor severity," *Tremor Other Hyperkinet. Mov.*, vol. 6, art. 375, 2016.

[28] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," *Proc. IEEE*, vol. 77, no. 2, pp. 257–286, 1989.

[29] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart rate variability: standards of measurement, physiological interpretation, and clinical use," *Circulation*, vol. 93, no. 5, pp. 1043–1065, 1996.

[30] R. Moe-Nilssen and J. L. Helbostad, "Estimation of gait cycle characteristics by trunk accelerometry," *J. Biomech.*, vol. 37, no. 1, pp. 121–126, 2004.

[31] O. Walch, Y. Huang, D. Forger, and C. Goldstein, "Sleep stage prediction with raw acceleration and photoplethysmography heart rate data derived from a consumer wearable device," *Sleep*, vol. 42, no. 12, 2019; data: O. Walch, "Motion and heart rate from a wrist-worn wearable and labeled sleep from polysomnography," PhysioNet, doi: 10.13026/hmhs-py35.

[32] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[33] R. Mattfeld, E. Jesch, and A. Hoover, "A new dataset for evaluating pedometer performance," in *Proc. IEEE Int. Conf. Bioinformatics Biomedicine (BIBM)*, 2017, doi: 10.1109/BIBM.2017.8217769.

[34] M. Karas *et al.*, "Labeled raw accelerometry data captured during walking, stair climbing and driving," PhysioNet, 2021, doi: 10.13026/51h0-a262.